

Bugs found: 4

1. Memory leak: the memory allocated for 'buf' using malloc is never released using free.
2. Use-after-free: the pointer 'dangling' is dereferenced and assigned a value after it has been freed.
3. Undefined behavior: left-shifting a 32-bit integer by 40 bits exceeds the width of the type.
4. Buffer overflow: the loop condition ' $i \leq 20$ ' causes an out-of-bounds write to 'buf[20]', as the buffer only has 20 bytes (indices 0-19).